

Cortical Organization in Shrews: Evidence From Five Species

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ABSTRACT

Cortical organization was examined in five shrew species. In three species, *Blarina brevicauda*, *Cryptotis parva*, and *Sorex palustris*, microelectrode recordings were made in cortex to determine the organization of sensory areas. Cortical recordings were then related to flattened sections of cortex processed for cytochrome oxidase or myelin to reveal architectural borders. An additional two species (*Sorex cinereus* and *Sorex longirostris*) with visible cortical subdivisions based on histology alone were analyzed without electrophysiological mapping. A single basic plan of cortical organization was found in shrews, consisting of a few clearly defined sensory areas located caudally in cortex. Two somatosensory areas contained complete representations of the contralateral body, corresponding to primary somatosensory cortex (S1) and secondary somatosensory cortex (S2). A small primary visual cortex (V1) was located closely adjacent to S1, whereas auditory cortex (A1) was located in extreme caudolateral cortex, partially encircled by S2. Areas did not overlap and had sharp, histochemically apparent and electrophysiologically defined borders. The adjacency of these areas suggests a complete absence of intervening higher level or association areas. Based on a previous study of corticospinal connections, a presumptive primary motor cortex (M1) was identified directly rostral to S1. Apparently, in shrews, the solution to having extremely little neocortex is to have only a few small cortical subdivisions. However, the small areas remain discrete, well organized, and functional. This cortical organization in shrews is likely a derived condition, because a wide range of extant mammals have a greater number of cortical subdivisions. *J. Comp. Neurol.* 410:55–72, 1999. © 1999 Wiley-Liss, Inc.

Indexing terms: insectivore; somatosensory; evolution; cortex; brain

Our primary interest in exploring the organization of neocortex in shrews relates to their small body size and, hence, their small brain size. We sought to address the basic question of how brain organization might change to compensate for extremes in size. Mammals vary greatly in body size, and, for the most part, mammalian brain size is correlated closely with body size (Jerison, 1973; Harvey and Krebs, 1990). Shrews include some of the smallest of mammals. The smallest shrews have tiny brains with little neocortex, the characteristic feature of mammalian brains. The smallest species of shrew in our study weighed only about 3 grams, whereas the largest mammals, blue whales, weigh 100 tons: roughly 30 million times more. The largest land mammals ever, the now extinct relatives of the modern rhinoceros, tipped the scales at 30 tons. Because there are major structural and functional consequences of such large size, 30 tons may represent the upper size limit for land mammals (Schmidt-Nielsen, 1984). There are certainly limits on the small end of the scale as well, and the smallest of shrews seem to be at or close to that limit.

We do not know all of the constraints on further size reduction in mammals; however, the need to pump blood and oxygenate a small body may be critical factors, especially given relatively high mammalian metabolic requirements (Schmidt-Nielsen, 1984). A shrew's heart pumps blood at a rate of up to 1,300 beats per minute, and faster rates may not be possible. Another possible constraint on size reduction may be the need to maintain certain brain functions.

Mammalian neocortex has a characteristic laminar organization (Brödmann, 1909), and this sheet of tissue contains a number of organ-like subdivisions that vary in both

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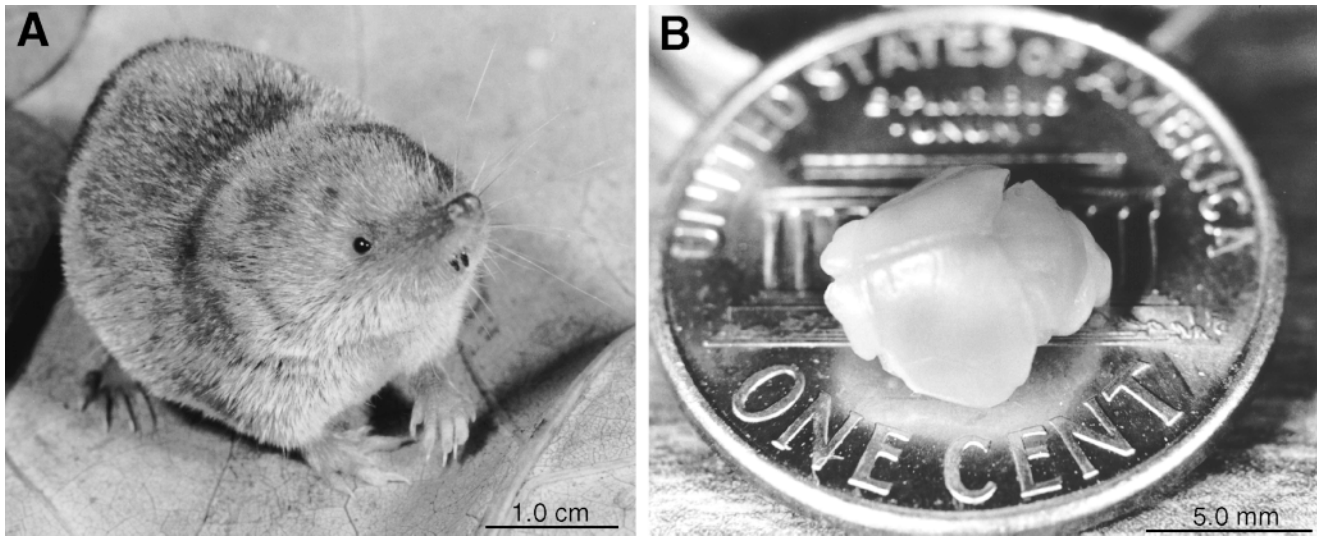


Fig. 1. The least shrew (*Cryptotis parva*) and an example of its relatively small brain. **A:** An adult least shrew (*C. parva*) investigating a leaf. This species typically weighs 4–5 grams and has small eyes and short, dense fur. The teeth are darkly pigmented as a result of iron deposits in the enamel. The snout is elongated and pointed and is

covered with sensory vibrissae, which are a major source of tactile information. These anatomic features are typical of soricine shrews, although the size and prominence of the eyes and ears vary across species. **B:** The brain of an adult least shrew resting on a penny for scale.

size and number with the total neocortex size (Kaas, 1987a,b, 1995; Northcutt and Kaas, 1995). Mammals with little neocortex must have either smaller areas, fewer areas, or both. Reducing the size of an area reduces its computational capacity, whereas losing an area removes its functions. How small can cortical areas be and still be able to perform their basic functions, and how many areas are really needed for mammals to act like mammals? Have shrews altered the basic design of neocortex to reduce the computational problems of having extremely small brains? These are questions we sought to answer by investigating cortical organization in shrews. The questions are especially interesting because we know that smaller brains cannot simply be scaled-down versions of larger brains. The basic computational element of brains, the neuron, does not scale up or down in size very well. Smaller brains have fewer neurons, and, even if these neurons function more efficiently in small brains, there is a computational price to pay when one reduces the number of neurons.

We also have other reasons for studying the organization of neocortex in shrews. Judging from fossil brain endocasts from the Triassic, the first mammals had small brains with little neocortex (Kielan-Jaworowska, 1983, 1984, 1986; Jerison, 1990). Many extant mammals also have small bodies (Gittleman and Purves, 1998) and brains (Jerison, 1973), and some of these species have been studied in efforts to gain insights into the organization of neocortex in early mammals (see, e.g., Krubitzer et al., 1986, 1993, 1997; Krubitzer, 1995; Beck et al., 1996). Shrews and other insectivores are especially relevant in this regard, because an insectivorous life style was common to many early eutherians. Although lineages remain somewhat uncertain, the roots of the insectivore order and of many other eutherian orders reach back to a time of divergence over 65 mya (Eisenberg, 1981; Novacek, 1992; de Jong, 1998). Because of descent and life style, modern insectivores may have retained many of the traits of early mammals.

Shrews represent over 70% of all insectivore species and are found throughout most of the world. They are mouse-like animals with short, dense fur, small eyes, and prominent sensory vibrissae (Fig. 1). Most feed on insects and other small invertebrates. Shrews are legendary for their high metabolisms, with some species consuming several times their body weight each day. These tiny mammals have been the subject of over 6,000 research papers, because their diversity, phylogenetic position, and wide distribution make them useful models for investigations of many aspects of mammalian biology (see Haberl, 1995). However, almost nothing is known about the organization of the sensory cortex. Early studies reported that attempts to record from the brains of some insectivores resulted in their immediate death (Allison and Van Twyver, 1970). However, improvements in microelectrode recording techniques and a better understanding of small mammal anesthesia have made it possible to make extensive recordings from the cortex of even small shrews. In this study, we used multiunit microelectrode recordings and histologic examination of flattened cortical preparations to determine the organization of sensory cortex in five species of shrew.

MATERIALS AND METHODS

Shrew species examined

For this study, cortical organization was examined in five species of shrew from three genera, which included the following species: *Sorex cinereus* (masked shrew), *Sorex longirostris* (southeastern shrew), *Sorex palustris* (northern water shrew), *Blarina brevicauda* (short-tailed shrew), and *Cryptotis parva* (least shrew). All species are found in North America and belong to the Soricinae subfamily (red-toothed shrews). Specimens were collected under Pennsylvania scientific collecting permit COL 00087 and Missouri permit 4566. In addition, specimens of *C. parva*

were used from a laboratory colony maintained at Kirksville College, Missouri (see Mock, 1982).

There was considerable variation in total brain and body size across the species examined. The smallest species, *S. longirostris*, had a body weight of only 3 grams, whereas the larger species, *B. brevicauda*, weighed from 15 grams to 20 grams. There also were differences in their susceptibility to stress. Whereas *C. parva*, the least shrew, is a hardy laboratory animal despite its diminutive size (4–5 grams), the two smallest species, *S. cinereus* and *S. longirostris*, both are highly sensitive to stress and can die from toe clipping or anesthetic injections. For this reason, only anatomical data were collected from these two species. For the remaining three species, multiunit electrophysiological recordings were used to map sensory cortex before the cortical anatomy was examined. Cortical recordings were made from 11 specimens of *C. parva*, 5 specimens of *B. brevicauda*, and 1 specimen of *S. palustris*. All experiments and animal care procedures were approved by the Vanderbilt University Animal Care Committee and were in compliance with the National Institutes of Health guidelines for the care and use of laboratory animals.

Cortical recording procedures

Shrews were anesthetized with a combination of urethane (15% weight/volume in distilled water) and ketamine (one-tenth diluted stock). This combination of anesthetics is not appropriate for survival surgeries. Shrews initially were given an intraperitoneal injection of 0.005 ml 15% urethane per gram of body weight (750 mg/kg). After 5–10 minutes, an additional intraperitoneal injection of 0.005 ml one-tenth diluted (in saline) stock ketamine (stock ketamine was 100 mg/ml) per gram of body weight was given (50 mg/kg). This usually was sufficient to obtain a deep surgical plane of anesthetic. Additional injections of one-fourth to half of the original ketamine dosage were given as a supplement when needed. Body temperature was maintained with a heating pad, which was separated from the animal by several layers of cloth to dampen the heat-transfer rate. An incision was made in the scalp, and the dorsal cranium was exposed. The dorsal side of the cranium contralateral to the recording hemisphere was then acrylated to a movable metal post. This procedure facilitated somatosensory and auditory stimulation, because most stereotaxic frames are too large to hold shrews and mechanically interfere with body stimulation used to define small tactile receptive fields. The cortex was exposed over one hemisphere and covered with silicon fluid without removing the dura. A photograph of the brain was made with Polaroid 55 negative film under a dissecting microscope with the aid of a flash unit. Multiunit microelectrode recordings were made in middle layers of cortex by using low-impedance Tungsten microelectrodes (1.0–1.5 M ohms at 1,000 Hz) with penetrations (through the dura) perpendicular to the cortical surface. Neuronal responses were amplified, viewed on an oscilloscope, and monitored on a speaker. Each electrode penetration site was marked on an enlarged photomicrograph of the brain, and selected penetration sites were lesioned (10 μ A for 10 seconds) to match the mapping results to sections of flattened cortex. Tactile receptive fields were defined under a surgical microscope by stimulating the skin of the nose and other body parts, deflecting vibrissae, and stimulating the hair on the body with small probes or a fine brush. Auditory stimuli included clicks and taps to metal bars and rubbing

two wooden probes together to produce a broad-spectrum, low-amplitude auditory stimulus. The best frequencies were not defined. Visual responses were obtained with small moving bars of light, full field flashes, or moving an object through a light beam to produce a moving shadow in the visual field. Visual receptive fields were not defined.

Recording sessions generally lasted 2–5 hours for the smallest species mapped (*C. parva*). This usually allowed time for about 30 electrode penetrations, although, in some cases, as many as 50 penetrations were mapped. This was sufficient to cover most of the small area of responsive cortex in some cases. Nevertheless, the best comprehensive map of *C. parva* was obtained by synthesizing the results from multiple experiments. In some cases, we concentrated on defining the extent of visually responsive cortex, whereas, in other cases, specific body parts or auditory cortex were mapped in greater detail. In the largest species, *B. brevicauda*, survival times were generally 5–8 hours and were sufficient to obtain 80–90 penetrations in individual cases. In the single specimen of *S. palustris* (northern water shrew), 25 electrode penetrations were mapped.

Cortical histology

After microelectrode mapping, shrews were given an overdose of sodium pentobarbital and perfused (with a 10-cc syringe) through the heart with saline followed by 2% paraformaldehyde in phosphate buffer, pH 7.4. The brain was removed, and the cortex from each hemisphere was separated from the underlying white matter. Cortex was left in 30% sucrose for 15 minutes, then flattened between glass slides, and stored in 30% sucrose for 4–24 hours before sectioning. To obtain sections parallel to the cortical surface, a sucrose block was cut to flatten its surface parallel to the knife edge, then the cortex was frozen to the block pia side down. Cortices were sectioned on a freezing microtome at 40–60 μ m. Sections were processed for either cytochrome oxidase (CO), Nissl, or myelin. For CO, sections were processed according to Wong-Riley and Carroll (1984) but were first incubated in 0.5% cobalt chloride (followed by a rinse in 0.05 M Tris buffer, pH 7.2) to increase the final contrast. Sections were stained for myelin by using the Gallyas method (Gallyas, 1979) and were stained for Nissl substance with thionin. Computer-generated photomicrographs were produced by scanning in photomicrographs to which digital labels were added subsequently. Only brightness and contrast were adjusted digitally. The sizes of different sensory areas in cortex were calculated from cases with borders visible in CO-processed sections.

RESULTS

The results of this study indicate a common plan of cortical organization in shrews. The five shrew species each revealed slightly different aspects of this organization, but the results were complementary, showing the same overall configuration of cortical areas. The most extensive maps from microelectrode recordings were obtained from the larger *B. brevicauda* specimens, whereas the most dramatic chemoarchitecture showing the same divisions within primary somatosensory cortex (S1) and distinctive borders for primary visual cortex (V1) were seen in the two smallest *Sorex* species from which recordings were not possible. The size and appearance of visual

cortex varied across the species examined, perhaps due to different degrees of dependence on vision (the *Sorex* species seemed to have the most prominent eyes) but always was found in the same relative location based on either recordings (*B. breviceauda* and *C. parva*) or CO staining (*S. cinereus*, *S. longirostris*, and *S. palustris*). Overall, the relative size of visual cortex seemed to be the most variable feature of cortex across the species, whereas the location of auditory cortex and the organization of somatosensory areas were similar in all species. Before describing the details of cortical organization, we briefly discuss the overall anatomy of the different shrews.

Some anatomical features of shrews in this study

Shrews make up the family Soricidae, one of six insectivore families. The Soricidae consists of 345 species divided into two subfamilies, the Soricinae and the Crocidurinae. All of the species in this study are in the Soricinae sub-family, which comprises most of the northern temperate species. Members of this subfamily typically have a higher metabolism and are characterized by darkly pigmented teeth (Fig. 1). Shrews in general have small eyes and prominent sensory vibrissae. Most authors conclude, based on the generally small size of the eyes and behavioral experiments, that vision is of little use for finding prey (Churchfield, 1990; Branis and Burda, 1994). Instead, shrews seem to depend heavily on smell, tactile cues, and auditory information as they search for food.

Shrews have a long, tapered snout covered with many sensory vibrissae and tipped with a small glabrous rhinarium. We sectioned the whisker pad and made counts of the vibrissae from *C. parva* and *B. breviceauda*, the two species from which extensive microelectrode recordings were obtained. Both whisker pads were organized similarly, with larger "macrovibrissae" located at the posterior dorsal area and smaller "microvibrissae" surrounding the mouth and in the anterior area just caudal to the rhinarium, which is typical for other mammal species (Brecht et al., 1997). There was a total of roughly 120 vibrissae in *B. breviceauda*, divided into 30 macrovibrissae and 90 microvibrissae. In the smaller *C. parva*, there was a total of 63 vibrissae, divided into 21 macrovibrissae and 42 microvibrissae. This difference in vibrissae number may be a result of the overall size difference between the two species (*C. parva* has a much smaller snout).

There is a single report (Pernetta, 1977) of the presence of Eimer's organs on the snout of a shrew (*Sorex minutus*). Eimer's organs are tactile sensory structures that have the form of an epidermal dome containing a series of sensory receptors (Catania, 1995). These organs are found on all talpid moles that have been examined and can be identified as 40–60 μm , domed epidermal swellings on the glabrous rhinarium. We found no external sign of this sensory organ on the rhinarium of the species in this study.

Results of cortical recordings

The recordings identified a relatively large portion of caudally located cortex that responded to tactile stimulation of the contralateral body surface. The results are summarized in Figure 2. There were two topographic representations of the body surface found within this area: one located rostromedially and the other located caudolaterally. In the more rostral area, the mouth was located laterally, followed by the face (vibrissae and glabrous

snout), and then the trunk and limbs in a roughly lateral-to-medial sequence. The second representation was a mirror image of the first reflected along the representation of the glabrous nose and the vibrissae. The trunk and limbs did not have adjoining representations but, rather, were separated by the much larger representations of the vibrissae. In the more caudal body representation, the mouth was located laterally, followed by the face, and then the trunk and limbs in a lateral-to-caudomedial sequence. No clear difference was found between the sizes of receptive fields in the two somatosensory areas.

In each species, auditory responses were recorded from a small area of cortex located caudal and lateral to the somatosensory area. Neurons at recording sites near the juncture of somatosensory and auditory cortex appeared to respond to either auditory or somatosensory stimulation, but not to both; however, the possibility of some overlap with somatosensory neurons also responding to auditory stimuli was not investigated systematically along the entire border.

Visual cortex was found just dorsal and caudal to the vibrissae representation and caudal to the representation of the trunk and limbs in the more dorsal of the two body representations. Here again, there was no apparent overlap between visual cortex and the somatosensory area.

Each of the different cortical areas identified with electrophysiology corresponded in location to the presence of histochemically apparent areas in the sections of flattened cortex. Both visual and auditory areas were coextensive with uniformly dark-staining divisions of cortex processed for either CO or myelin. The somatosensory areas also stained darkly for both CO and myelin, but they were heterogeneous in CO preparations, containing a series of light-staining septa or different patterns of staining corresponding either to divisions between areas or to representations of specific body parts, respectively. Specific details from individual species are presented below.

Recordings from *B. breviceauda*

Electrophysiological recordings were made from cortex of five specimens of *B. breviceauda*, two of which are illustrated in Figures 3 and 4. Sensory areas were located relatively caudally, occupying much of the caudal half of cortex above the rhinal fissure. In case 97-81 (Fig. 3), we mapped the somatosensory areas in detail, much of the extent of auditory cortex, and a single penetration (Fig. 3, point 44; marked with a star to indicate a lesion) that responded strongly to vision. In case 97-47 (Fig. 4), only somatosensory responses were mapped. The most obvious result, which held true in every species examined, was that the vibrissae representation dominated the somatosensory areas. In case 97-81, the vibrissae occupied roughly 60% of cortex responsive to tactile stimulation. When the estimated areas of both auditory cortex and visual cortex were added into the total, the facial vibrissae occupied roughly 50% of all sensory cortex identified in this experiment (for more detailed maps of visual cortex, see Table 1 and the results discussed below). Clearly, these sensory organs are of great importance to shrews, which is predictable from their peripheral anatomy and behavior. An island of cortex embedded within the vibrissae representation was identified that responded to the glabrous tip of the snout (Figs. 3, 4). Neurons at the rostralateral extreme of the somatosensory areas responded to stimulation of the lips, teeth, and mouth. In sections processed for CO or

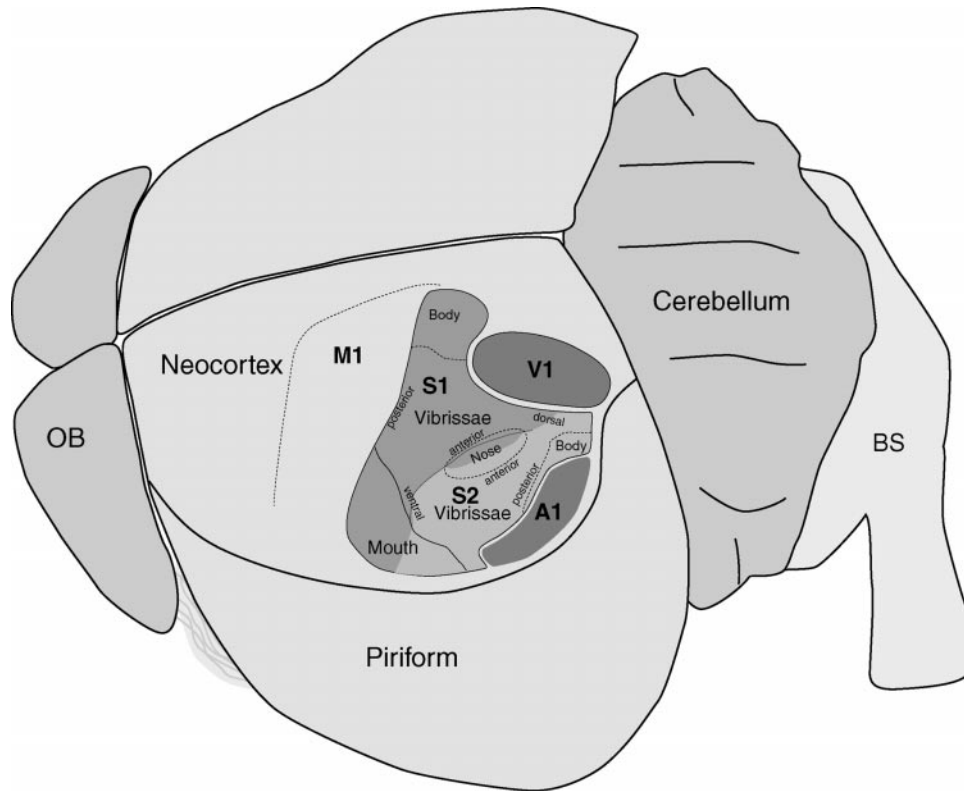


Fig. 2. Summary of cortical organization in shrews. The sensory areas were located in caudal, lateral cortex. Two representations of the contralateral body surface were found, one corresponding in location and organization to the primary somatosensory area (S1), and the other more lateral area corresponding in location and organization to the second somatosensory area (S2). Auditory cortex (A1) was found at the extreme caudolateral end of neocortex, partially surrounded by S2. We tentatively identify this area as A1 based on its myeloarchitecture. Visual cortex was found just dorsal and caudal to S1 and varied in size across the species examined (it is depicted here as it was found in the

more visual *Sorex* species; see text and Figs. 11, 12). A striking feature of cortical organization in these small-brained mammals is the adjacency of sensory areas. Both electrophysiologic and histologic evidence indicated clear, sharp borders between areas that often were separated by less than 200 μm . This condition seems to exclude the possibility of intervening higher level or association areas typical of other larger brained mammals. The location and extent of motor cortex was estimated based on a previous study of corticospinal projections in *C. parva* (see text and Fig. 14). OB, olfactory bulb; BS, brainstem; V1, primary visual cortex; M1, presumptive motor cortex.

myelin, a light septa usually separated the mouth representation from the rest of somatosensory cortex (see the cases discussed below).

Two representations of the trunk and limbs were identified: one at the medial extreme of somatosensory cortex and another at the caudal extreme. For example, in case 97-81 (Fig. 3), neurons responded to stimulation of the forelimb medially at penetration 72 and caudolaterally at penetrations 54 and 55. A similar topography was found in case 97-47 (Fig. 4) for penetrations 47 and 4.

There also was strong evidence for two representations of the vibrissae: one located on each side of the representation of the glabrous snout. For example, in Figure 3, penetrations 19 and 53 were located at opposite extremes of the somatosensory representation yet respond to the same vibrissae. Other examples from the same case include penetrations 20, 36, 91, and 31 as well as penetrations 16 and 37. In case 97-47 (Fig. 4), similar examples include penetrations 40 and 25 as well as penetrations 33 and 43. In general, it was difficult to obtain responses from single vibrissae; instead, most penetrations responded to small groups of vibrissae. This could be due to the small absolute size of the somatosensory representations in shrews, causing the representations of different body parts

to crowd closer together and reducing the resolution of multiunit recordings. Nevertheless, the overall topography of the two vibrissae representations was determined and was consistent across cases. In both vibrissae representations, the ventral vibrissae on the snout were located laterally in cortex, closest to the cortical representation of the mouth, whereas the dorsal vibrissae on the snout were represented at the opposite, dorsocaudal extreme of somatosensory cortex. For the other dimension, anterior vibrissae on the snout were located caudally in the more rostral vibrissae representation and rostrally in the caudal vibrissae representation, such that, in both cases, the representations of the most anterior vibrissae on the snout generally were found adjacent to the representation of the glabrous rhinarium in the cortex. This organization matches the overall topography of the peripheral skin surface, because anterior vibrissae on the snout are closest to the glabrous rhinarium.

The most posterior vibrissae on the snout were represented at the dorsorostral extreme in the rostral cortical representation and at the caudal and dorsal extreme in the caudal cortical representation. Again, this seemed to match the overall topography of the skin surface, because caudal vibrissae on the snout (those located closest to the trunk)

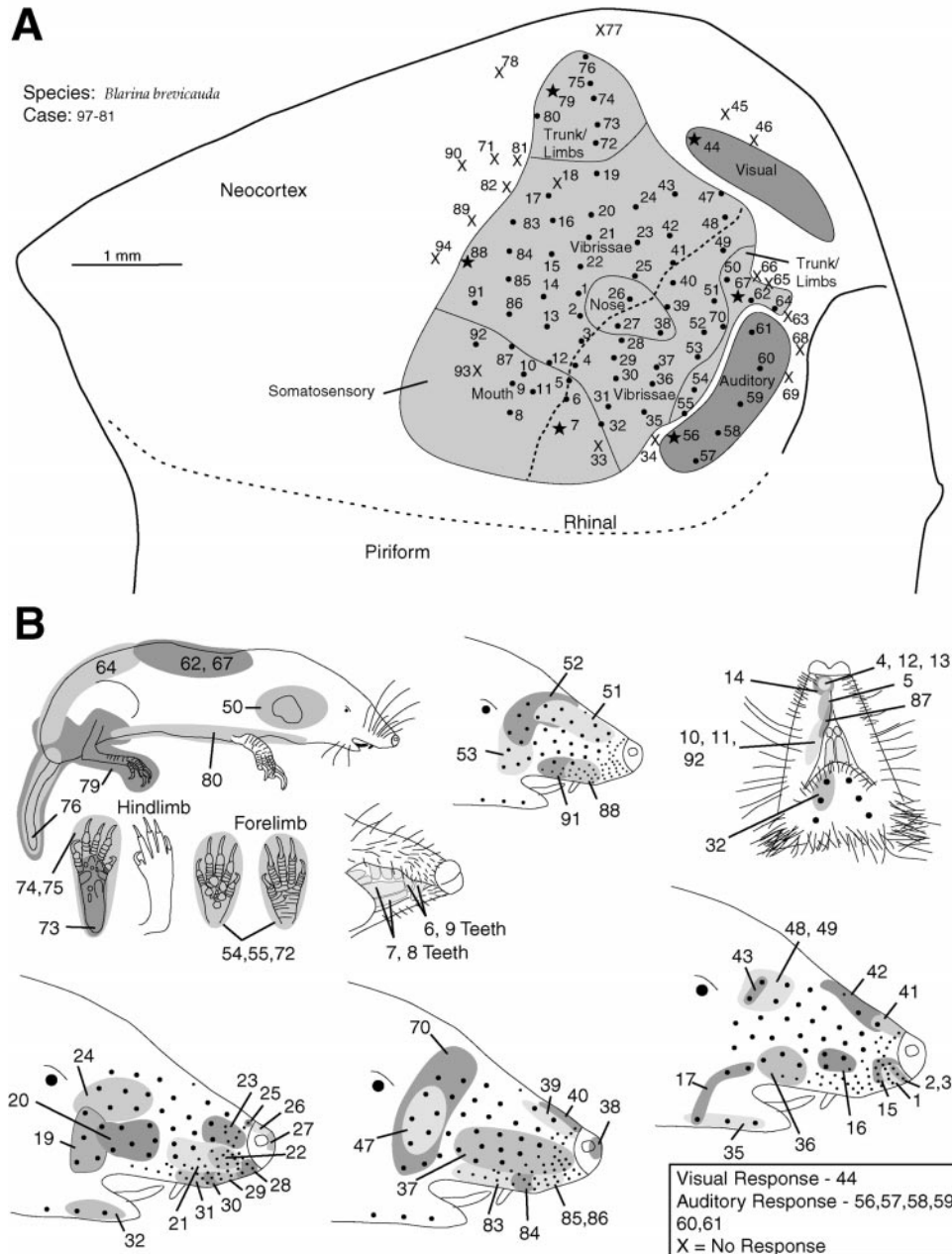


Fig. 3. Microelectrode recording results from *Blarina brevicauda*, the short-tailed shrew. **A:** Drawing of the electrode penetrations in correlation to a section of flattened cortex processed for cytochrome oxidase (CO) to reveal chemoarchitectonic borders (shaded areas). **B:** The receptive fields for the electrode penetrations indicated in A. In this case (97-81), somatosensory cortex was mapped in detail as well as most of the extent of auditory cortex. In the somatosensory region, the mouth was found at the rostralateral extreme. Just dorsal and caudal to the mouth, a large area of cortex responded to the contralateral vibrissae. Within the vibrissae representation, a small, central island of cortex responded to the glabrous snout (nose). Two representations of the trunk and limbs were identified: one at the medial extreme of responsive cortex and the other at the caudal extreme, just rostral and medial to auditory cortex. Auditory cortex formed a crescent just above the rhinal sulcus. A single penetration (44)

responded to visual stimuli. Within the extensive vibrissae representation, ventral vibrissae on the snout were represented closest to the mouth, whereas caudal vibrissae on the snout were represented at the opposite caudomedial extreme, closest to visual cortex and the body representations. Anterior vibrissae on the snout were generally represented closest to the glabrous rhinarium, whereas posterior vibrissae on the snout were represented farthest from glabrous rhinarium, near auditory cortex caudolaterally, or adjacent to the visual area and the medial representation of the trunk and limbs. This organization indicated the presence of two vibrissae representations: one just dorsomedial to the glabrous rhinarium and the other on the opposite caudolateral side. In A, the dashed line is a rough division between S1 rostrally and S2 caudally. In B, large dots represent single macrovibrissae, and small dots represent two microvibrissae. Stars mark the locations of microlesions.

generally were represented closest to the trunk and limb representations in cortex. This configuration of cortex indicated two mirror image representations of the vibrissae.

Where appropriate, details were mapped. Our best estimate of the division between the two representations is indicated by a dashed line in Figures 3 and 4 (for B.

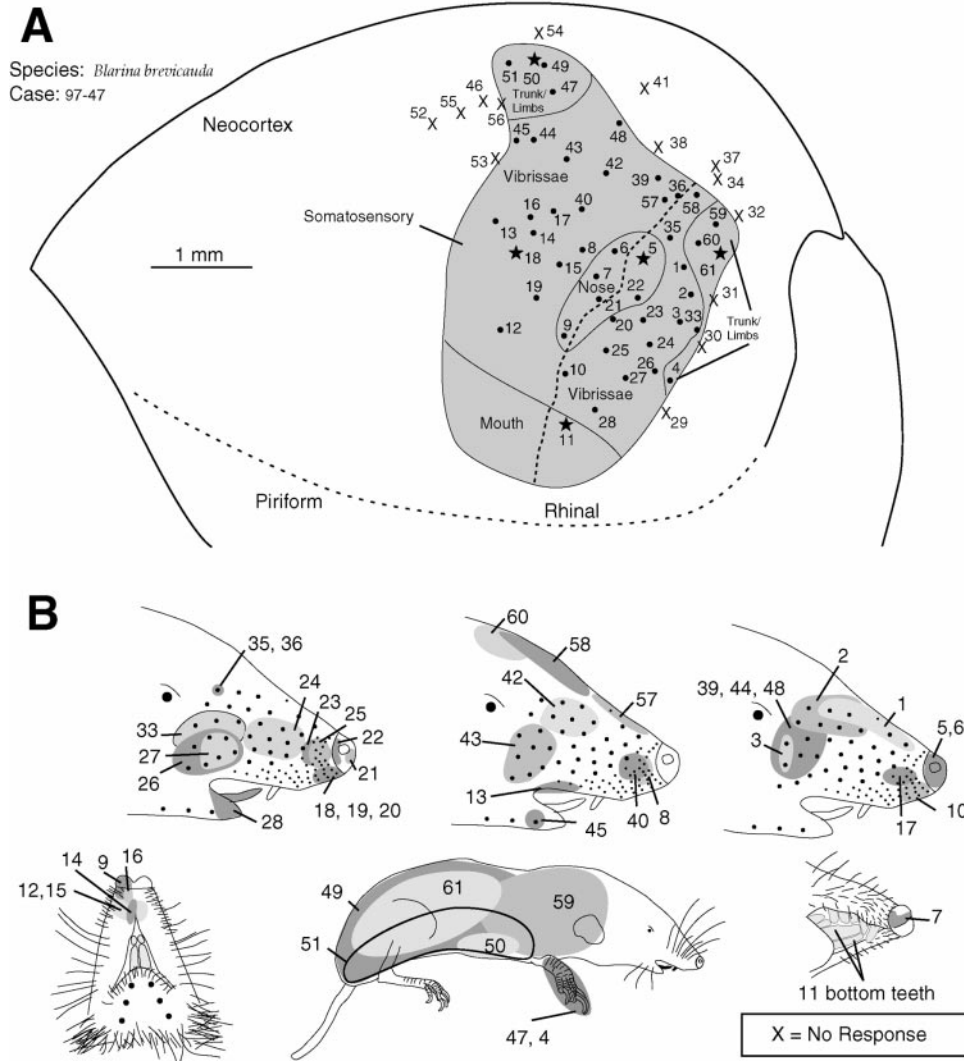


Fig. 4. Microelectrode recording results from a second *B. brevicauda* (case 97-47). **A:** Drawing of the electrode penetrations in relationship to a section of flattened cortex processed for CO to reveal chemoarchitectonic borders (shaded areas). **B:** The receptive fields for the electrode penetrations indicated in A. In this case, only somatosensory responses were mapped. The organization was similar to the previous case, showing the lateral-to-medial topographic representation of the body in two representations (again, divided into S1 and S2

roughly along the dashed line). Trunk and limbs were identified both medially (penetrations 47, 49, 50, and 51) and caudally (penetrations 4, 59, 60, and 61). Two vibrissae representations were indicated, one on either side of the glabrous rhinarium, whereas the mouth was at the lateral extreme (penetration 11). In B, large dots represent single macrovibrissae, and small dots represent two microvibrissae. Stars mark the locations of microlesions.

TABLE 1. Sizes of Neocortical Areas (in mm²)

Cortical area ¹	Sorex cinereus (%)			Blarina brevicauda (%)			Cryptopus parva (%)
	Case 1	Case 2A	Case 2B	Case 97-81	Case 97-82	Case 97-43	Case 97-44
Total neocortex	15.0 (100)	15.0 (100)	14.8 (100)	29.6 (100)	27.9 (100)	26.9 (100)	21.8 (100)
Total sensory cortex (S1, S2, A1, V1)	5.0 (33)	5.0 (33)	4.7 (32)	9.8 (33)	8.8 (32)	8.8 (33)	7.5 (34)
A1	0.46 (3.1)	0.44 (2.9)	0.42 (2.8)	0.86 (2.9)	0.97 (3.5)	0.87 (3.2)	0.30 (1.4)
V1	0.84 (5.6)	0.76 (5.1)	0.69 (4.7)	0.64 (2.2)	0.56 (2.0)	0.41 (1.5)	0.61 (2.8)
Total somatosensory cortex (S1 + S2)	3.7 (25)	3.5 (23)	3.6 (24)	8.3 (28)	7.3 (26)	7.5 (28)	6.6 (30)
Vibrissae and rhinarium (S1 + S2)	2.6 (17)	2.4 (16)	2.6 (18)	5.0 (17)	—	—	—
Trunk and limbs (S1 + S2)	0.45 (3.0)	0.46 (3.1)	0.39 (2.6)	1.3 (4.4)	—	—	—
Mouth	0.60 (4.0)	0.72 (4.8)	0.65 (4.4)	2.0 (6.8)	—	—	—

¹The sizes of the neocortical areas in mm² were estimated from chemoarchitectonic subdivisions combined, in some cases, with electrophysiological recordings. Areas were measured after fixation in 2% paraformaldehyde. A1, auditory cortex; S1, primary somatosensory cortex; S2, secondary somatosensory cortex; V1, primary visual cortex.

brevicauda). This division also is supported by the chemoarchitecture of flattened cortex (see histologic results below). The dorsalmost representation is appropriately

organized and located for S1, the primary somatosensory representation, whereas the caudolateral representation has the appropriate organization and location for the

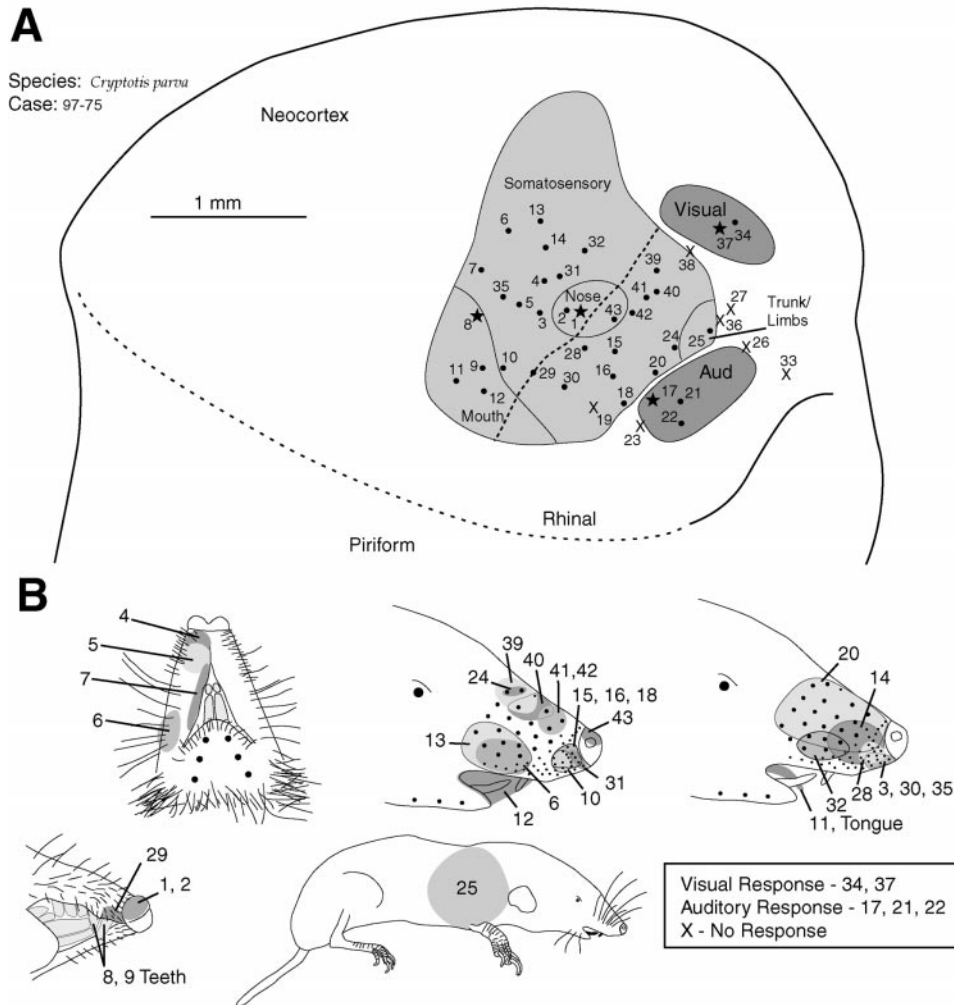


Fig. 5. Recording results from *C. parva*, the least shrew (case 97-75). **A**: Drawing of the electrode penetrations in correlation to a section of flattened cortex processed for CO to reveal chemoarchitectonic borders (shaded areas). **B**: The receptive fields for the electrode penetrations indicated in A. Shorter survival times under anesthesia resulted in slightly fewer penetrations within single cases for this species. However, details from many cases provided evidence for the overall organization of sensory areas. In this case, much of the face in somatosensory cortex was mapped as well as points in visual and auditory cortex. The organization was very similar to that of *B.*

brevicauda. Results suggest two representations of the vibrissae: one on either side of the glabrous nose (penetrations 1, 2, and 43). The mouth was located laterally (penetrations 8, 9, 11, and 12). In some cases, very closely spaced electrode penetrations responded to separate sensory modalities (penetration 17 vs. penetrations 18 and 20). This indicated sharp borders between areas (for distinct, histologically apparent subdivisions, see also Figs. 11 and 12). In B, large dots represent single macrovibrissae, and small dots represent single microvibrissae. Stars mark the locations of microlesions.

secondary somatosensory area (S2; see Discussion). There was not sufficient resolution to identify duplicate representations of the glabrous snout because of its small external size and cortical representation or, in the case of the mouth representation, due to difficulties of systematically mapping small, internally located structures. However, based on previous studies, it seems justified to presume that these representations likewise are divided into S1 and S2 areas along a congruent border (see, e.g., Catania and Kaas, 1995, 1997). Strong auditory responses were obtained in a strip of cortex at the caudolateral extreme, just above the rhinal sulcus.

Recordings from *C. parva*

Cortical recordings were made from 11 specimens of *C. parva*, 5 of which are illustrated in Figures 5–9. The

organization of sensory areas was highly similar to *B. brevicauda*, with the mouth located rostrolaterally, followed by vibrissae and body representations in a lateral-to-medial sequence. Some additional body parts were identified; for example, responses were obtained from the tongue representation in case 97-75 (Fig. 5, penetration 11) and from the hind limb representation in presumptive S2 in case 97-80 (Fig. 7, penetration 25). Although there were significantly fewer vibrissae on the face of *C. parva*, the vibrissae representation still dominated the sensory areas. Widely separated electrode penetrations responded to vibrissae, such as penetrations 6 and 20 in case 97-75 (Fig. 5) or penetrations 16 and 8 in case 97-79 (Fig. 6). Repeated receptive fields indicated two vibrissae representations, which also were seen in *B. brevicauda*. In case 97-80 (Fig. 7), the dorsal-to-lateral progression of penetrations 5–17

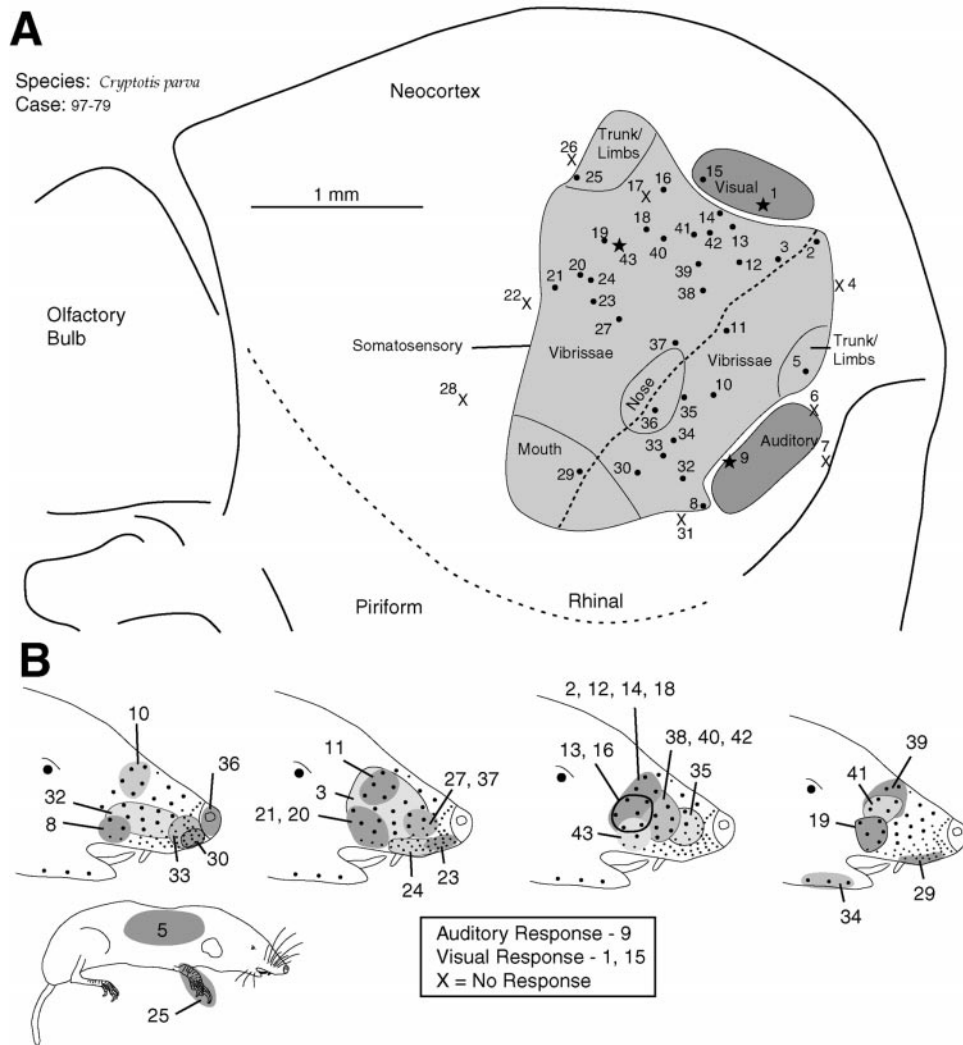


Fig. 6. Recording results from *C. parva*, the least shrew. **A:** Drawing of the electrode penetrations in correlation to a section of flattened cortex processed for CO to reveal chemoarchitectonic borders (shaded areas). **B:** The receptive fields for the electrode penetrations indicated in A. In this case (97-79), again, much of the somatosensory cortex was mapped as well as points in visual and auditory cortex. Results here also suggest two representations of the vibrissae: one on either side of the glabrous nose (penetration 36). The mouth was located laterally, and two representations of the trunk and limbs were

identified in the same relative location as in *B. brevicauda*. In some cases, closely spaced electrode penetrations responded to separate sensory modalities (penetration 14 vs. penetrations 1 and 15; penetration 9 vs. penetrations 8 and 32). This indicated sharp borders between areas and was supported by the histology, which showed subdivisions with septa along borders of areas. In B, large dots represent single macrovibrissae, and small dots represent single microvibrissae. Stars mark the locations of microlesions.

in cortex followed a dorsal-to-ventral progression of receptive fields for the vibrissae, which also was found in S2 of *B. brevicauda*. Also, like *B. brevicauda*, the glabrous rhinarium had a relatively small representation as an island of cortex encircled by the two vibrissae representations (Figs. 5, 6).

Because recording experiments were shorter in this species, different sensory areas were mapped most extensively in individual cases. In cases 97-75 and 97-79 (Figs. 5, 6), somatosensory areas were mapped in detail. In case 97-80 (Fig. 7), much of the extent of auditory cortex was determined, whereas, in case 97-77 (Fig. 8), only visual responses were recorded. Overall, the results indicated a large, caudally located somatosensory area dominated by the vibrissae representation. Auditory and visual areas

were located caudolaterally and dorsocaudally, respectively, directly adjacent to the somatosensory areas, as in *B. brevicauda*. The electrophysiological responses indicated sharp transitions between regions of cortex responsive to auditory, visual, or tactile stimulation, although the possibility of some bimodal cortex was not fully eliminated. In some cases, penetrations that responded exclusively to either auditory or tactile stimulation were found less than 200 μm apart (Fig. 5, penetrations 18, 20, and 17). This also was typical for visual and tactile responses, and, in some cases, penetrations responding to either vision or tactile stimulation were less than 150 μm apart (Fig. 7, penetrations 4 and 5). Similar results were found for *B. brevicauda* for auditory cortex (Fig. 3, penetrations 35, 55, and 56).

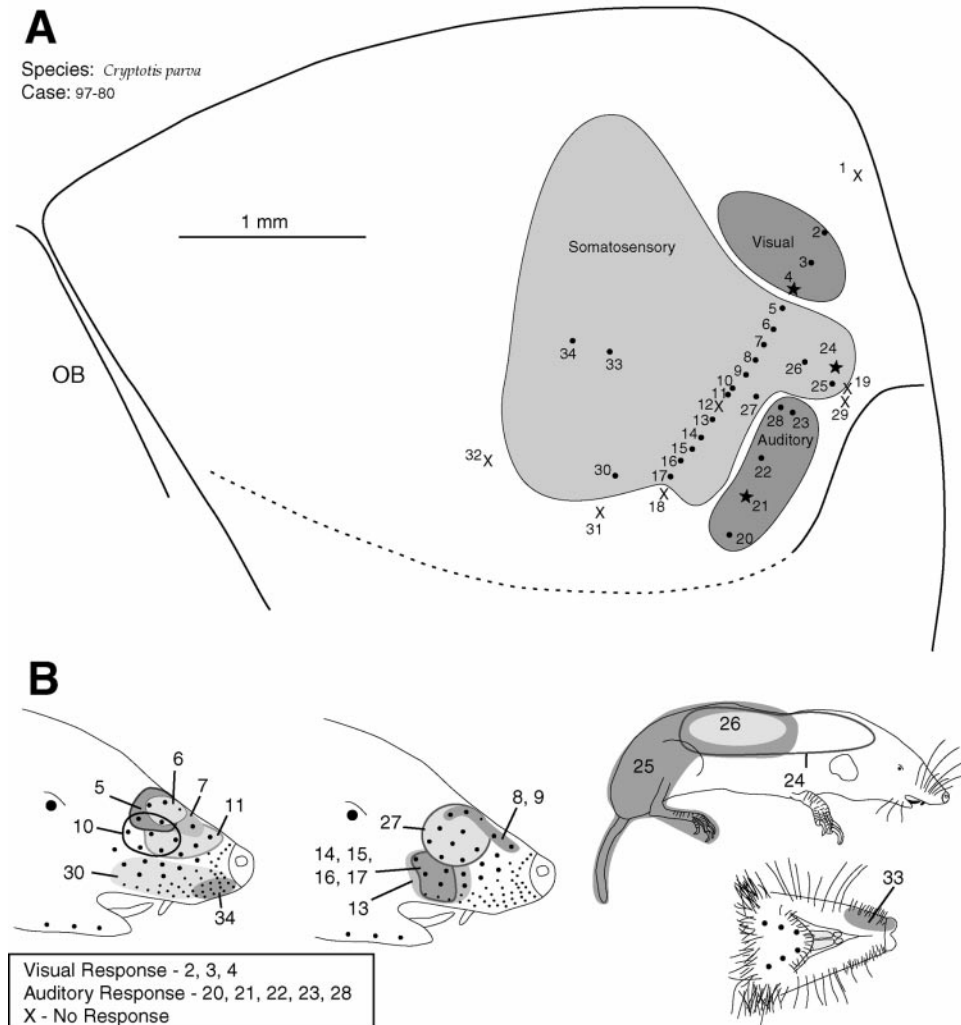


Fig. 7. Recording results from *C. parva* (case 97-80). **A**: Drawing of the electrode penetrations in correlation to a section of flattened cortex processed for CO to reveal chemoarchitectonic borders (shaded areas). **B**: The receptive fields for the electrode penetrations indicated in A. In this case, a medial-to-lateral progression of electrode penetrations shows a dorsal-to-ventral progression of receptive fields in presump-

tive S2. Note the sudden transition between visual and somatosensory responses; penetrations 4 and 5 are roughly 140 μ m apart (adjusted for fixative shrinkage). The S2 hindlimb representation was identified in caudal cortex (penetration 25). Auditory cortex was just caudal to S2. In B, large dots represent single macrovibrissae, and small dots represent single microvibrissae. Stars mark the locations of microlesions.

Although the recordings are not extensive, case 97-44 (Fig. 9) is illustrated to show a matching cortical section stained for CO that contained many of the chemoarchitectural borders and microlesions within a single section. In this case, most of V1, a presumptive primary auditory cortex (A1), and the large, darkly staining somatosensory region can be discerned. The two vibrissae representations are located within most of the extent of the large CO-dense region rostral to auditory cortex and rostralateral to visual cortex. The mouth representation is separated from the vibrissae representation by a light septum (Fig. 9, arrow-head). In the dorsal region of the vibrissae area, the darkest staining region of cortex is located appropriately for the S1 vibrissae representation. Although individual barrels could not be identified, this area is distinctly patchy.

Recordings from *S. palustris*

Recordings were made from the cortex of a single specimen of *S. palustris*, the water shrew. The map of cortex consists of 25 recording sites examined for tactile responses only. The results are complementary to the findings in the other species, revealing a large expanse of cortex responsive primarily to the sensory vibrissae (Fig. 10). Responses from the trunk were obtained from areas in medial cortex and from the presumptive S1 representation, and a single penetration was responsive to the glabrous snout (Fig. 10, penetration 25), surrounded by the vibrissae representations, as in the other species examined. Some evidence for two vibrissae representations was found, indicated by similar receptive fields mapped for different somatosensory areas (Fig. 10, penetra-

Species: *Cryptotis parva*
Case: 97-77

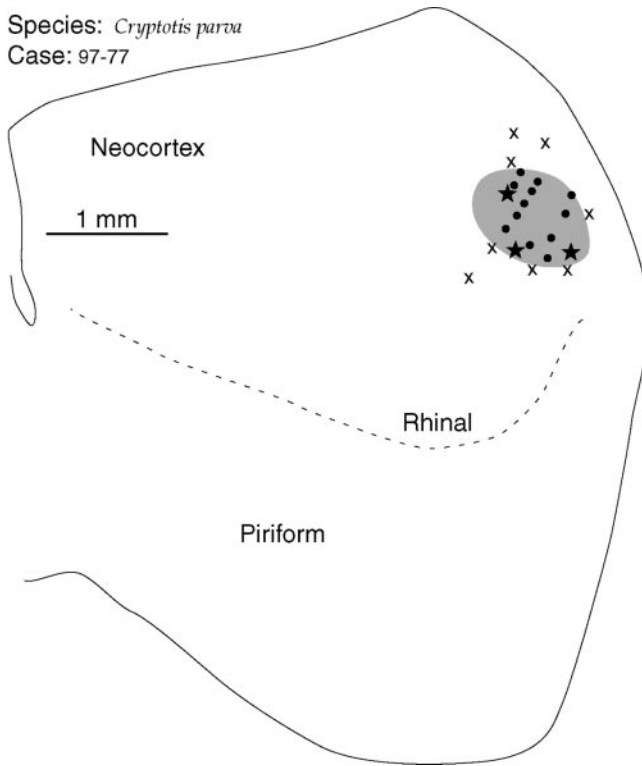


Fig. 8. A map of the extent of visually responsive cortex in *C. parva*. In this case, only visual responses were mapped in a series of closely spaced microelectrode penetrations. Fourteen penetrations in a small oval of cortex responded to full-field light, moving bars and circles, flashes, or dark bars passed through full-field light (off responses). These points were flanked by penetrations that did not respond to visual stimuli. The location and size of the area correspond to a uniform CO-dark or myelin-dense region of cortex just dorsal and caudal to the S1 vibrissae representation (see Figs. 9, 10, 12). Stars mark the locations of microlesions.

tions 14 and 23). Although neither auditory nor visual cortex was mapped, sections processed for CO revealed dense staining regions of cortex that corresponded to the locations of visual cortex and auditory cortex identified in the recordings from other species.

CO staining from *S. cinereus* and *S. longirostris*

The two small *Sorex* species were highly sensitive to stress; therefore, we did not attempt recordings from cortex. However, flattened brain sections processed for CO or myelin revealed distinctive fields and borders that corresponded in location to the cortical areas identified by recordings and histology in the other three species. Only the CO-processed sections are presented, because they showed the greatest detail. Figure 11 shows two flattened sections from a single hemisphere of *S. cinereus*. These sections illustrate the size and location of the sensory areas relative to the entire flattened cortex. Similar to the other species, the areas are located in caudal cortex. In Figure 11A, the primary somatosensory representation of the facial vibrissae is apparent as a patchy, dense CO region in the middle area of cortex. A light septum delineates the border between the vibrissae and the rostralat-

eral mouth representation (Fig. 11A, arrowhead), and a separate light septum is found laterally, presumably separating the S1 and S2 vibrissae representations in the vicinity of the glabrous nose area. There is an obvious change in chemoarchitecture along this S1/S2 border. Auditory cortex is located in caudal, lateral cortex, indicated by a uniformly dark crescent of tissue. In an adjacent section (Fig. 11B), visual cortex is clearly apparent as a uniform, dark oval in a position just dorsal and caudal to the S1 vibrissae representation. The location of visual cortex, as indicated by CO activity, is in the same relative location that was found in the electrophysiological map for *C. parva* (Fig. 8). In a second specimen of *S. cinereus* (Fig. 12) the same areas are apparent in a single flattened section of cortex, showing the direct adjacency of the different sensory areas. Once again, somatosensory, auditory, and visual areas each are delineated by characteristic features of the tissue processed for CO, with subdivisions identifying some body parts.

Finally, a single specimen of *S. longirostris* was examined (Fig. 13), and the reconstruction of sensory areas from multiple sections is included to demonstrate the similarity of cortical organization across the species examined. The basic organization of cortex, based on histology, was similar to that of *S. cinereus* and the other species.

Motor cortex

In a previous study of corticospinal neurons, Nudo and Masterton (1990) identified the locations of cells projecting to the spinal cord in *C. parva*. We were able to align the foci of these previously identified cells with the sensory areas and cortical borders from the present study (Fig. 14); thus, we were able to identify the likely location of primary motor cortex (Fig. 2). The focus was located medially in cortex, just rostral to the primary somatosensory representation of the body. Because only neurons that project to the spinal cord were labeled, the location of the cranial motor representation was not identified but, presumably, would be found in more lateral cortex.

Surface area of neocortex and sensory representations

An important attribute of the species examined in this study was their small body size and correspondingly small brains with relatively little neocortical area. Thus, the results would not be complete without a report of the overall size of neocortex and its sensory subdivisions, at least for the cases that were mapped in detail or revealed sufficient resolution of areas based on cortical histology (Table 1).

The largest species examined was *B. brevicauda*, which generally weighed between 15 grams and 20 grams. The total neocortical area, estimated from fixed, flattened cortical sections, is shown in Table 1 for three cases that had particularly clear cortical subdivisions. These three cases had an average of 28 mm² of neocortex, of which approximately 9 mm² (32%) were devoted to combined somatosensory, visual, and auditory areas. For case 97-81, enough detail was obtained from the extensive microelectrode recordings and histologic borders to estimate the subdivisions of somatosensory areas and the borders of V1 and A1 as well. Both V1 and A1 occupied less than 1 mm² each. V1 took up approximately 0.6 mm², whereas A1 occupied roughly 0.9 mm². The somatosensory area occu-

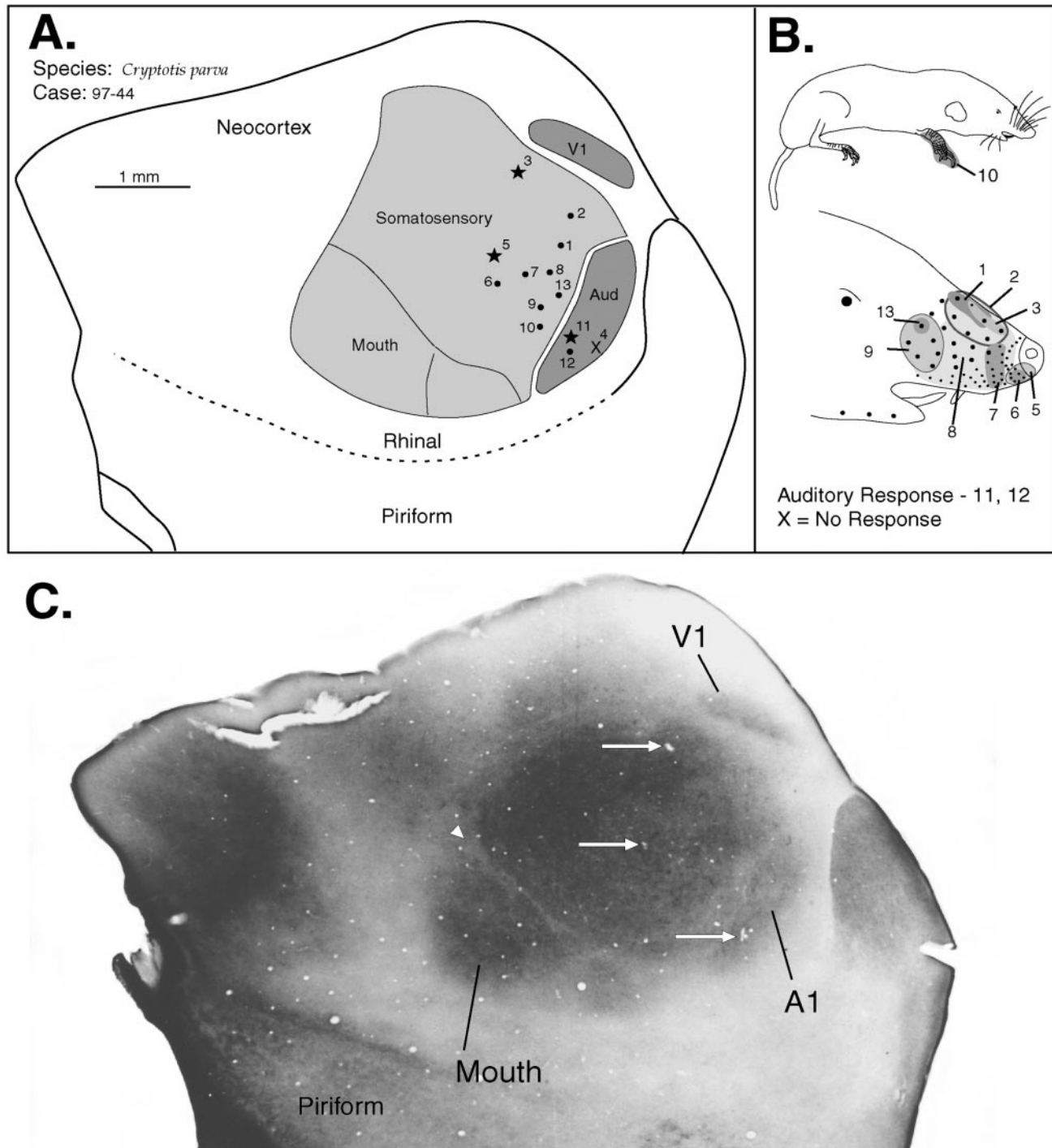


Fig. 9. Recording results from *C. parva* (case 97-44). **A:** Drawing of the electrode penetrations in correlation to a section of flattened cortex processed for CO. **B:** The receptive fields for the electrode penetrations indicated in A. **C:** A section of flattened cortex processed for CO to reveal the subdivisions of cortex. The large, dark staining region in caudal cortex contains the somatosensory areas and is dominated by the vibrissae representations. Just dorsal to the somatosensory area, a small stripe of cortex is located in an area appropriate for the primary

visual cortex (V1). Caudal and lateral to the somatosensory areas, a small crescent of dark tissue marks the extent of auditory cortex (A1). The mouth is represented at the anterior and lateral extreme, separated from the vibrissae by a CO light septa (arrowhead). Arrows mark the locations of microlesions in C. Stars mark the corresponding electrode penetrations in A. In B, large dots represent single macrovibrissae, and small dots represent single microvibrissae.

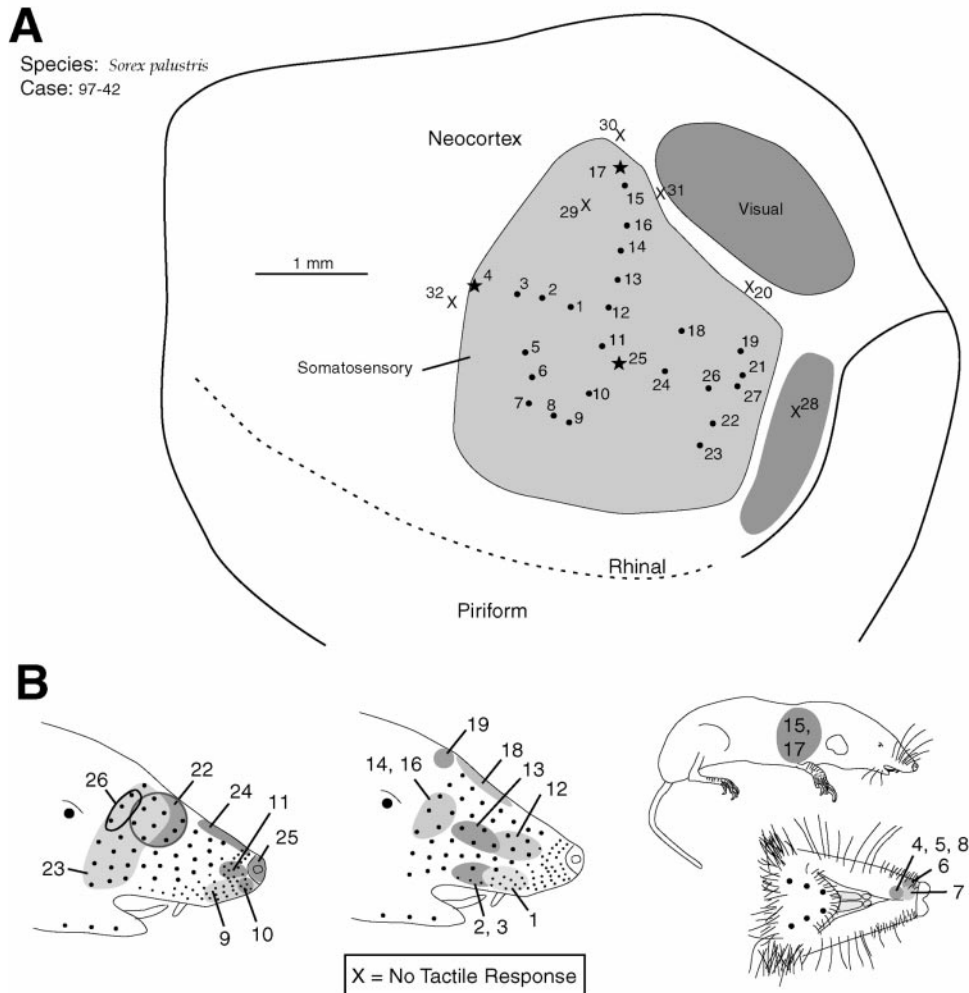


Fig. 10. Recording results from the water shrew, *Sorex palustris* (case 97-42). **A:** Drawing of the electrode penetrations in correlation to a section of flattened cortex processed for CO to reveal chemoarchitectonic borders (shaded areas). **B:** The receptive fields for the electrode penetrations indicated in A. Although recordings were not extensive in this species, the results show similarities to the other species, includ-

ing evidence for the medially located body representation in S1 (penetrations 15 and 17), a glabrous rhinarium representation surrounded by vibrissae (penetration 25), and some evidence of two vibrissae representations, as indicated by widely separated electrode penetrations with similar receptive fields (penetrations 14 and 16 vs. penetration 23). Stars mark the locations of microlesions.

ing roughly 8.3 mm², dominating sensory cortex. The majority of somatosensory cortex in both S1 and S2 was taken up by the representation of the vibrissae.

In the case of the smaller species of *S. cinereus*, neocortical area was estimated for three hemispheres from two specimens. The average area of cortex for these cases was a mere 15 mm², of which 5 mm² were occupied by sensory areas (S1, S2, V1, and A1). It is striking that, despite the great difference in total neocortical area between *B. brevicauda* and *S. cinereus*, sensory areas in both species occupied one-third of neocortex. Although microelectrode mapping was not possible in this species, the chemoarchitecture and differential CO staining of cortical areas allowed us to estimate the subdivisions of cortex. Similar to *B. brevicauda*, somatosensory cortex dominated the sensory areas and, itself, was dominated by the representation of the vibrissae.

DISCUSSION

We were interested in defining subdivisions of neocortex in shrews, because some species of shrews have among the smallest brains of extant mammals (Hofman, 1985; Stephan et al., 1991), and early mammals were shrew-like in appearance and life style (Colbert and Morales, 1991). Thus, the brains of shrews offer opportunities to learn about adjustments to small brain size and about brain evolution in early mammals. By using microelectrode mapping procedures and comparing results with cortical architecture in the same shrews, we obtained clear evidence for two somatosensory representations, S1 and S2, an auditory field (presumably A1), and a primary visual area (V1). A previous description of the location of corticospinal projection neurons in shrews (Nudo and Masterton, 1990) provides evidence for a primary motor field just rostral to S1 (Fig. 1). The organization of S1 in shrews

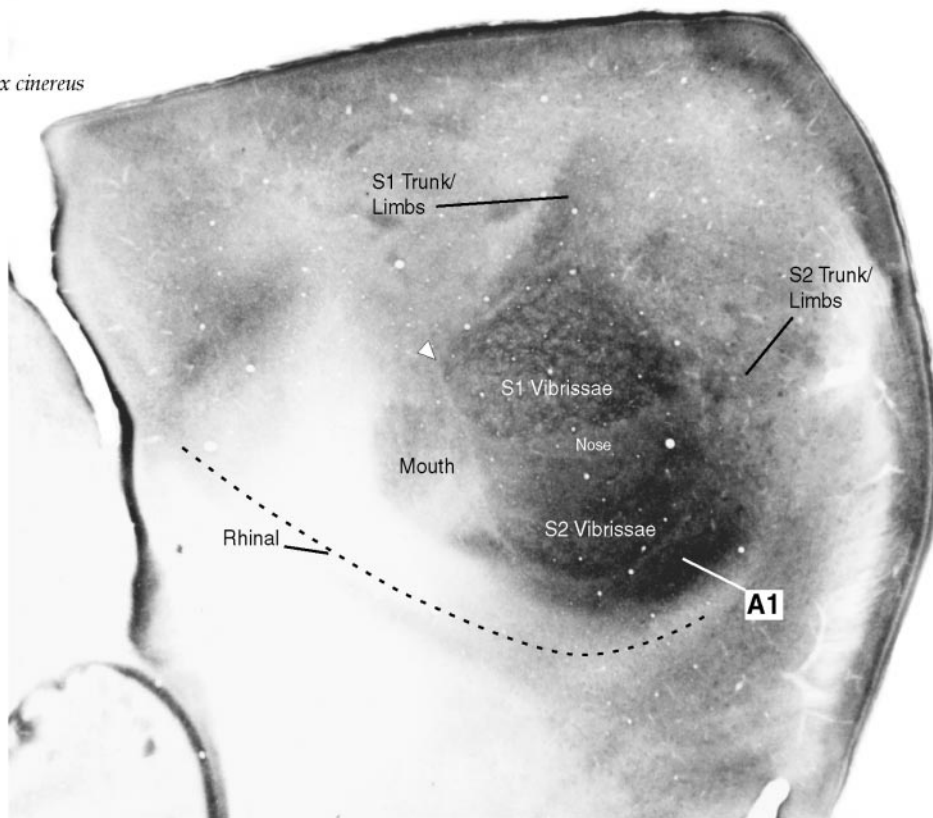
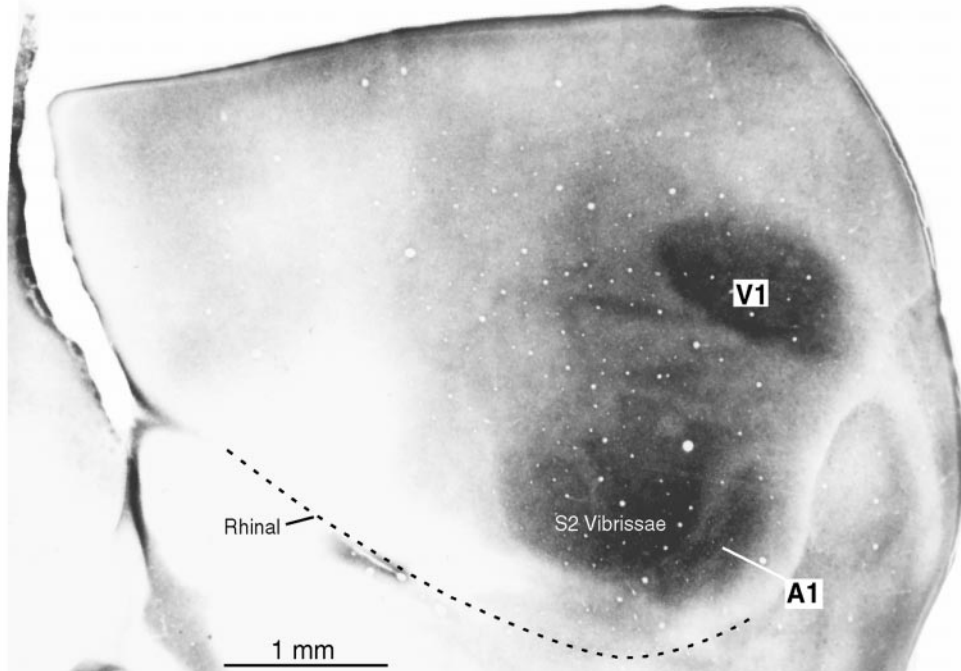
A.Species: *Sorex cinereus***B.**

Fig. 11. Tangential sections through the cortex of *Sorex cinereus* showing cortical areas and borders revealed by distinctive CO histochemistry. **A,B:** Sections showing the organization of cortex. The visible areas are similar in location and organization to areas identified by electrophysiology and histology in the other species, although, here, visual cortex is relatively larger and more distinct. In A, the location of the primary somatosensory cortex (S1) vibrissae is evident from the patchy, CO-dark tissue in middle cortex. The secondary somatosensory cortex (S2) vibrissae representation is just lateral to S1 and has a more uniform appearance. The mouth is located directly

rostral and slightly lateral to the vibrissae, separated by a CO-light septum (arrowhead). Auditory cortex (A1) is located just posterior and lateral to S2 (marked by a dark crescent of tissue). In B, the location and extent of visual cortex are illustrated dramatically by a CO-dense oval of tissue in the middle posterior cortex (V1), whereas auditory cortex, likewise, is stained darkly. Note the direct adjacency of the different cortical sensory areas. The dashed lines indicate the location of the rhinal sulcus. This is illustrated for a single section of cortex in Figure 12.

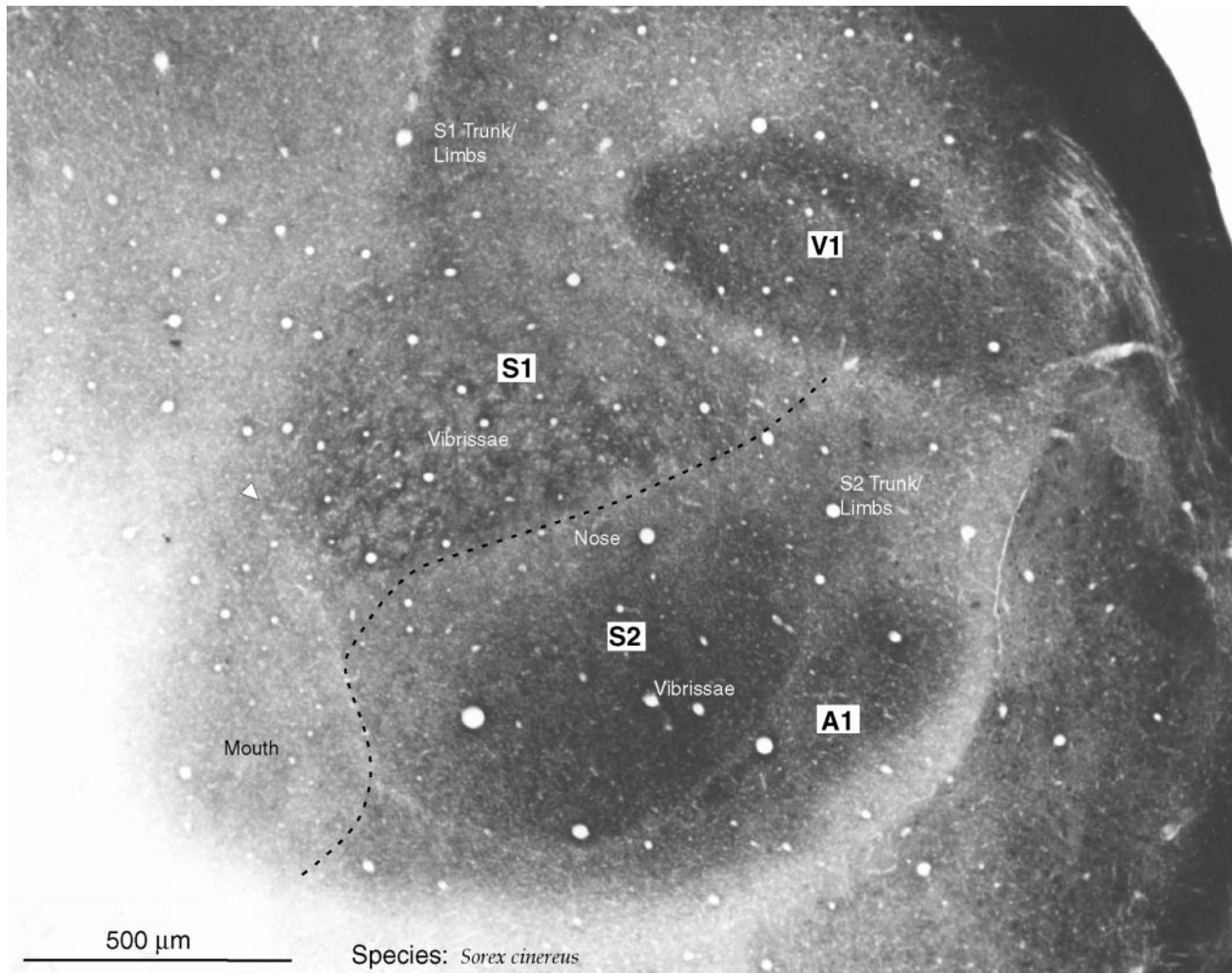


Fig. 12. A section of cortex from *S. cinereus* demonstrating the direct adjacency of cortical areas in a single section processed for CO. For example, note the close proximity of the S1 vibrissae representation to V1 medially and the direct adjacency of A1 to S2. Each cortical

area has a distinctive appearance, whereas subareas (such as mouth, vibrissae, and trunk) within S1 and S2 also are distinct. The dashed line indicates the presumptive border between S1 and S2.

shares a common somatotopic pattern for small mammals (see Johnson, 1990), in that the body is represented medially, and the face and mouth are represented laterally, with much of the total area devoted to facial vibrissae, lips, and teeth. The secondary area, S2, also has a characteristic organization, mirroring S1, with the body caudolateral and the face farthest lateral. Shrews do not appear to have somatosensory fields other than S1 and S2.

The somatosensory and presumed motor fields occupy an extreme caudolateral position in neocortex that is unusual for many small mammals. Somatosensory areas, especially S2, are in the caudal half of neocortex, and they extend nearly to the caudal pole. A small auditory area at the caudal pole nearly reaches the rhinal fissure, whereas a small, laterally placed visual area borders S1 and S2 and the caudal tip of the rhinal fissure. In contrast, auditory and somatosensory areas are placed more rostrally in rats and mice (Johnson, 1990; Li et al., 1990), opossums (Beck et al., 1996), and insectivore tenrecs (Krubitzer et al., 1997) and hedgehogs (Kaas et al., 1970). However, the

arrangement of cortical fields in shrews is similar to that of closely related moles, in which cortical areas also are located caudally (Catania and Kaas, 1995, 1997).

The results indicate that the number of sensory and presumed motor fields in shrews is extremely limited, quite possibly fewer than in any other mammal. Most often, mammals have some cortex, frequently considerable amounts, between primary somatosensory, visual, and auditory areas. In shrews, the secondary somatosensory area, S2, occupies all of this space, primary visual and auditory areas border S1 and S2, and there is only a narrow rim of perirhinal cortex lateral and caudal to the sensory field. There is some cortex medial and rostral to the sensory and motor fields, and much of this cortex, judging from other mammals, is expected to be limbic and orbitofrontal cortex, which is important in the regulation of innate behaviors and motivational states (Kolb, 1990; Vogt et al., 1992). Tenrecs appear to have a visuomotor field rostral to M1 (Krubitzer et al., 1997), and rats may have a secondary or supplementary motor area rostral to

Species: *Sorex longirostris*

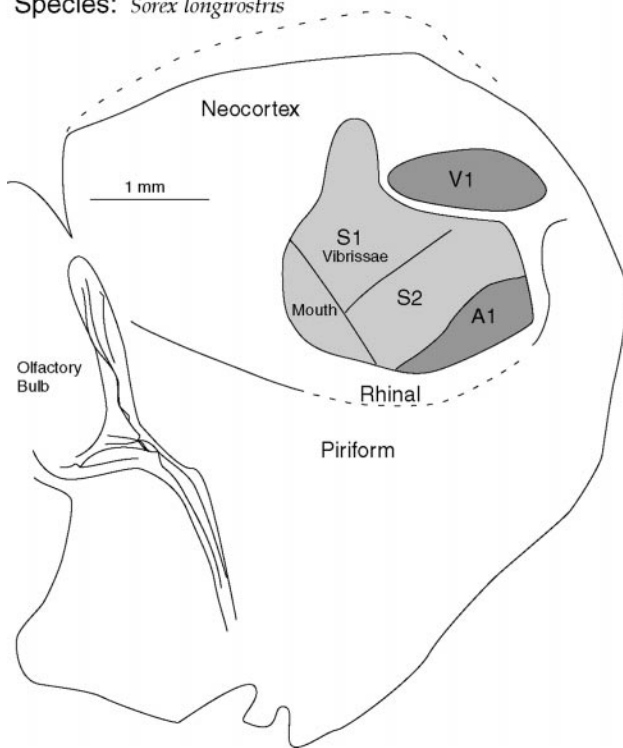


Fig. 13. The cortex and sensory areas in *Sorex longirostris* reconstructed from multiple sections of flattened and sectioned cortex that were processed for CO. The organization of cortex is similar to that of *S. cinereus* and the other shrews in this study.

M1 (Neafsey et al., 1986; Li et al., 1990). Thus, some cortex rostral to M1 in shrews may be sensorimotor. Nevertheless, present evidence suggests that shrews function with one auditory area, two somatosensory areas, and perhaps only one visual area—fewer areas than any other well-studied mammal. If one compares shrews with rats, for example, we see that rats have a granular S1 cortex, a surrounding dysgranular field that appears to function as another somatosensory field, and S2 and a parietal ventral area (PV), for a total of at least four somatosensory areas (Li et al., 1990). Auditory cortex in rats and mice includes A1 (Sally and Kelly, 1988) and several additional auditory fields (Stiebler et al., 1997), and visual cortex includes a number of extrastriate areas (see, e.g., Montero, 1993), even if one postulates a single area, V2, along the lateral border of V1 (Kaas et al., 1989). In addition to these fields, there is an expanse of temporal cortex that is not occupied by any of the known sensory areas. This temporal cortex most probably is involved in sensory processing, and it may include several sensory fields.

The major functional implication of the small numbers of sensory areas in shrews is that they are limited to rather simple and general-purpose sensory processing. A single visual area, by necessity, has to be involved in all cortical processing of visual information. If having multiple visual areas allows each field to specialize for some functions while giving up others, then it seems that little specialization is possible in shrew visual cortex. The complex visual systems of higher primates with 30 or more visual areas (Kaas, 1997) allows areas to become specialized for differ-

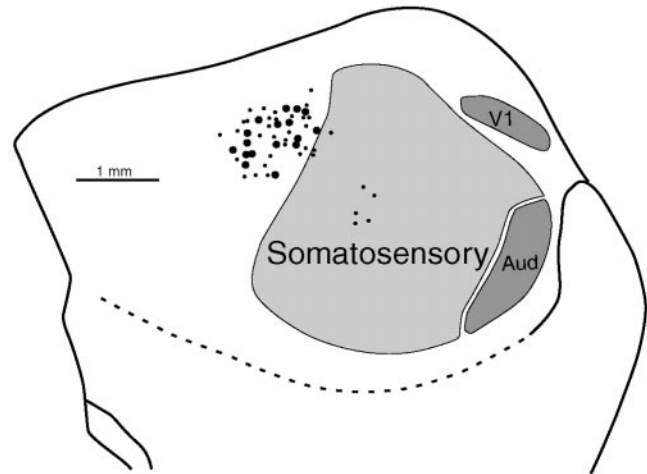


Fig. 14. The location of a focus of corticospinal projecting cells (large and small dots) estimated from data in Figure 8 of Nudo and Masterton (1990) and superimposed on the sensory areas in *C. parva*. Cells were identified in the original study by applying wheat germ agglutinin-horseradish peroxidase to the hemisected spinal cord of *C. parva* and processing the cortex later with tetramethyl benzidine to identify transported tracer. Here, the sensory areas are drawn based on the recordings and cytochrome oxidase sections from *C. parva* in the present study. The relative positions of the sensory areas and labeled cells were determined by aligning the cortical borders between the two cases.

ent functions; thus, great sophistication and range in visual behavior can be achieved. Although not much is known about visual behavior in shrews, we expect that such behaviors are rather simple. In addition, we expect that the response properties of neurons in visual cortex reflect the need for general purpose rather than specific functions. For example, visual cortex should connect directly with motor and limbic fields that are important in behavior, because, presumably, there are no higher order areas to do so, and neurons should be broadly tuned to stimulus parameters that affect behavior.

The small sizes of visual and auditory areas in shrews also suggest that neurons within these fields are highly interconnected with other neurons within the field and with a larger proportion of thalamic inputs. The proportional connectivity of neuronal structures decreases with increasing size (Ringo, 1991). This generally means that receptive fields would be large and would have limited spatial or tonotopic specificity. In shrews, somatosensory areas S1 and S2 both are considerably larger than V1 and A1, and neurons in both fields have reasonably restricted receptive fields. Clearly, the cortex of shrews emphasizes somatosensory processing, and we would expect more behaviors, especially those requiring some precise localization, to depend on the somatosensory modality.

Although the first mammals were shrew-like in appearance and life style, we feel that important features of the neocortical organization in shrews are specializations rather than retentions of an ancestral state. It is true that the evidence from fossil endocasts of the brain of early mammals suggests that they had small brains with little neocortex (Jerison, 1990), but the comparative evidence from other extant mammals, such as opossums (Beck et al., 1996), hedgehogs (Kaas et al., 1970), tenrecs (Krubitzer et al., 1997), and even the platypus (Krubitzer et al.,

1995), suggest that several more sensory areas existed in early mammals than in extant shrews and that these areas commonly have been retained in nearly all mammals. In particular, a secondary visual area (V2) and additional visual areas in temporal cortex lateral to V2 frequently have been described (see, e.g., Kaas, 1995), and it is reasonable to suppose that ancestral mammals had more visual areas than the one retained by shrews. Some mammals with poor vision, such as echolocating bats (Krubitzer, 1995), some moles (Catania and Kaas, 1995), and blind mole rats (Cooper et al., 1993a,b), appear to have reduced their commitment to visual processing, having only a single, small visual area that may retain few functions. However, shrews do not have a solely fossorial or cave-dwelling life style that would limit the usefulness of vision. Similar arguments can be made for the costs of limited cortical, auditory, and (to a lesser extent) somatosensory systems in shrews (see also Gould, 1969; Buchler, 1976). If early mammals had more sensory cortex and more sensory fields, then why would there be less in shrews?

We believe that shrews, at least the smaller species, are at the limits of small body size for mammals. Proportional metabolic costs skyrocket as body size decreases for homeothermic mammals (Schmidt-Nielsen, 1984). The brain is metabolically expensive tissue and contributes significantly to these metabolic costs (Williams and Herrup, 1988). One way that shrews may have adapted to small size is by reducing brain size and brain metabolism, especially by limiting sensory processing cortex (see also Yaskin, 1984, 1994). To some extent, this could be done by reducing the sizes of cortical areas; however, cortical areas probably need a minimal size and minimal numbers of neurons to perform even a few basic functions (Williams and Herrup, 1988). Thus, a maximal reduction in cortex might be possible only by eliminating some cortical areas. Shrews apparently have limited visual and auditory processing to single cortical areas. This may have been accomplished as an evolutionary adaptation to minimal body size.

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